

HOMEWORK 10 SOLUTIONS

SECTION 6.2

1 a) $\mathcal{U} = \{f(x) \mid f(x) \in P_3, f(2) = 1\}$ is not a subspace of P_3 :

$0(x) \notin \mathcal{U}$ because $0(2) = 0 \neq 1$.

b) $\mathcal{U} = \{xg(x) \mid g(x) \in P_2\}$ is a subspace:

i) $0(x) \in \mathcal{U}$ because $0(x) = x \cdot 0(x)$ with $0(x) \in P_2$.

ii) Let $p(x), q(x) \in \mathcal{U}$. Is $(p+q)(x) \in \mathcal{U}$?

$$\left. \begin{array}{l} p(x) \in \mathcal{U} \Rightarrow p(x) = xg(x) \text{ with } g(x) \in P_2 \\ q(x) \in \mathcal{U} \Rightarrow q(x) = xh(x) \text{ with } h(x) \in P_2 \end{array} \right\} \mapsto (p+q)(x) = p(x) + q(x) = xg(x) + xh(x) = x(\underbrace{g(x) + h(x)}_{\in P_2}) \Rightarrow (p+q)(x) \in \mathcal{U}.$$

iii) Let $p(x) \in \mathcal{U}, a \in \mathbb{R}$. Is $(ap)(x) \in \mathcal{U}$?

$$p(x) \in \mathcal{U} \Rightarrow p(x) = xg(x) \text{ with } g(x) \in P_2 \Rightarrow (ap)(x) = ap(x) = axg(x) = x(\underbrace{ag(x)}_{\in P_2}) \Rightarrow (ap)(x) \in \mathcal{U}.$$

e) $\mathcal{U} = \{p(x) \mid p(x) \in P_3, \text{ the constant term of } p(x) \text{ is } 0\} = \{ax + bx^2 + cx^3 \mid a, b, c \in \mathbb{R}\} = \text{span}\{x, x^2, x^3\}$.

So $\mathcal{U}$ is a vector subspace because it is the span of $\{x, x^2, x^3\}$.

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c) $\mathcal{U} = \{f \in F[0,1] \mid f(0) = f(1)\}$ is a vector subspace:

i) $0(x) \in \mathcal{U}$ because $0(0) = 0(1) = 0$.

ii) Let $f, g \in \mathcal{U}$. Is $f+g \in \mathcal{U}$?

$$\left. \begin{array}{l} f \in \mathcal{U} \Rightarrow f(0) = f(1) \\ g \in \mathcal{U} \Rightarrow g(0) = g(1) \end{array} \right\} \mapsto (f+g)(0) = f(0) + g(0) = f(1) + g(1) = (f+g)(1) \Rightarrow f+g \in \mathcal{U}.$$

iii) Let $f \in \mathcal{U}, a \in \mathbb{R}$. Is $af \in \mathcal{U}$?

$$f \in \mathcal{U} \Rightarrow f(0) = f(1) \Rightarrow (af)(0) = a f(0) = a f(1) = (af)(1) \Rightarrow af \in \mathcal{U}.$$

d) $\mathcal{U} = \{f \in F[0,1] \mid f(x) \geq 0 \text{ for all } x \in [0,1]\}$ is not a subspace because it is not closed under scalar multiplication:

$f(x) = 1$ is in $\mathcal{U}$, but $(-1) \cdot f(x) = -1 \cdot 1 = -1$ is not in $\mathcal{U}$.

g) $\mathcal{U} = \{f \mid f \text{ is integrable and } \int_0^1 f(x) dx = 0\}$ is a vector subspace:

i) $0(x) \in \mathcal{U}$ because $\int_0^1 0 dx = 0$

ii) Let $f, g \in \mathcal{U}$. Is $f+g \in \mathcal{U}$?

$$\left. \begin{array}{l} f \in \mathcal{U} \Rightarrow \int_0^1 f(x) dx = 0 \\ g \in \mathcal{U} \Rightarrow \int_0^1 g(x) dx = 0 \end{array} \right\} \Rightarrow \int_0^1 (f(x)+g(x)) dx = \int_0^1 f(x) dx + \int_0^1 g(x) dx = 0+0=0 \Rightarrow f+g \in \mathcal{U}.$$

iii) Let $f \in \mathcal{U}$, $a \in \mathbb{R}$. Is $af \in \mathcal{U}$?

$$f \in \mathcal{U} \Rightarrow \int_0^1 f(x) dx = 0 \Rightarrow \int_0^1 af(x) dx = a \int_0^1 f(x) dx = a \cdot 0 = 0 \Rightarrow af \in \mathcal{U}.$$

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b) x is in $\text{span}\{x^2+1, x+2\} \Leftrightarrow x = a(x^2+1) + b(x+2)$ for some $a, b \in \mathbb{R}$.

$$x = a(x^2+1) + b(x+2) = ax^2 + bx + (a+2b) \Leftrightarrow \begin{cases} a = 0 \\ b = 1 \\ a+2b = 0 \end{cases} \quad (\text{This system is inconsistent})$$

Therefore, $x \notin \text{span}\{x^2+1, x+2\}$

c) $\begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix}$ is in $\text{span}\left\{\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}\right\} \Leftrightarrow \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix} = a \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} + b \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$ for some $a, b \in \mathbb{R}$.

$$\begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix} = a \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} + b \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} a+2b & b-a \\ 2a+b & a \end{bmatrix} \Leftrightarrow \begin{cases} a+2b = 1 \\ b-a = 3 \\ 2a+b = -1 \\ a = 1 \end{cases} \quad (\text{Inconsistent})$$

Thus, $\begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix} \notin \text{span}\left\{\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}\right\}$.

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We have to show $\text{span}\{\vec{u}, \vec{v}, \vec{w}\} \subseteq \text{span}\{\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w}\}$ and $\text{span}\{\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w}\} \subseteq \text{span}\{\vec{u}, \vec{v}, \vec{w}\}$.

i) $\text{span}\{\vec{u}, \vec{v}, \vec{w}\} \subseteq \text{span}\{\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w}\}$?

$$\left. \begin{array}{l} \vec{u} = (\vec{u}+\vec{w}) - \vec{w} \in \text{span}\{\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w}\} \\ \vec{v} = -(\vec{u}-\vec{v}) + (\vec{u}+\vec{w}) - \vec{w} \in \text{span}\{\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w}\} \\ \vec{w} \in \text{span}\{\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w}\} \end{array} \right\} \xRightarrow{\text{Span theorem}} \text{span}\{\vec{u}, \vec{v}, \vec{w}\} \subseteq \text{span}\{\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w}\}.$$

ii) $\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w} \in \text{span}\{\vec{u}, \vec{v}, \vec{w}\}$ (they are linear combinations of $\vec{u}, \vec{v}, \vec{w}$) $\Rightarrow \text{span}\{\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w}\} \subseteq \text{span}\{\vec{u}, \vec{v}, \vec{w}\}$.

We conclude that $\text{span}\{\vec{u}, \vec{v}, \vec{w}\} = \text{span}\{\vec{u}-\vec{v}, \vec{u}+\vec{w}, \vec{w}\}$

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We have to prove $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n, \vec{0}\} \subseteq \text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$ and $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n, \vec{0}\} \supseteq \text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$.

i) $\vec{v}_1, \dots, \vec{v}_n \in \{\vec{v}_1, \dots, \vec{v}_n\} \subseteq \text{span}\{\vec{v}_1, \dots, \vec{v}_n\}$

$\vec{0} = 0\vec{v}_1 + \dots + 0\vec{v}_n \in \text{span}\{\vec{v}_1, \dots, \vec{v}_n\}$

$\Rightarrow \text{span}\{\vec{v}_1, \dots, \vec{v}_n, \vec{0}\} \subseteq \text{span}\{\vec{v}_1, \dots, \vec{v}_n\}$

ii) $\vec{v}_1, \dots, \vec{v}_n \in \{\vec{v}_1, \dots, \vec{v}_n, \vec{0}\} \subseteq \text{span}\{\vec{v}_1, \dots, \vec{v}_n, \vec{0}\} \Rightarrow \text{span}\{\vec{v}_1, \dots, \vec{v}_n\} \subseteq \text{span}\{\vec{v}_1, \dots, \vec{v}_n, \vec{0}\}$

$\text{span}\{\vec{v}_1, \dots, \vec{v}_n, \vec{0}\} = \text{span}\{\vec{v}_1, \dots, \vec{v}_n\}$

SECTION 6.3

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b) $\{x^2, x+1, 1-x-x^2\}$ is independent:

$$ax^2 + b(x+1) + c(1-x-x^2) = 0 \Rightarrow (a-c)x^2 + (b-c)x + (b+c) = 0 \Rightarrow \begin{cases} a-c=0 \\ b-c=0 \\ b+c=0 \end{cases} \Rightarrow a=b=c=0$$

Gaussian elimination

d) $\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is independent:

$$a \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + d \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a+c+d & a+b+d \\ a+b+c & b+c+d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} a+c+d=0 \\ a+b+d=0 \\ a+b+c=0 \\ b+c+d=0 \end{cases} \Rightarrow a=b=c=d=0$$

Gaussian elimination

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c) Is $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}$ a basis of M_{22} ? Since this set has 4 elements and $\dim M_{22}=4$, it is enough to show that the set is independent.

$$a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + d \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a+c+d & b+c \\ b & a+c \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} a+c+d=0 \\ b+c=0 \\ b=0 \\ a+c=0 \end{cases} \Rightarrow a=b=c=d=0$$

Gaussian elimination

We conclude that $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}$ is independent, so it is a basis of M_{22} because $\dim M_{22}=4$.

d) Is $\{1+x, x+x^2, x^2+x^3, x^3\}$ a basis of P_3 ? Because $\dim P_3=4$, it is enough to show linear independence.

$$a(1+x) + b(x+x^2) + c(x^2+x^3) + dx^3 = 0 \Rightarrow a + (a+b)x + (b+c)x^2 + dx^3 = 0 \Rightarrow \begin{cases} a=0 \\ a+b=0 \\ b+c=0 \\ c+d=0 \end{cases} \Rightarrow a=b=c=d=0$$

Gaussian elimination

Therefore, $\{1+x, x+x^2, x^2+x^3, x^3\}$ is independent, so it is a basis of P_3 since $\dim P_3=4$.

6c

$\mathcal{U} = \{p(x) \mid p(1) = 0\} \subseteq P_2$. We want to find a basis for $\mathcal{U}$.

Let $p(x) = a + bx + cx^2 \in P_2$. Then

$$p(x) \in \mathcal{U} \Leftrightarrow p(1) = a + b + c = 0 \Leftrightarrow p(x) = (-b-c) + bx + cx^2 = b(x-1) + c(x^2-1).$$

This yields $\mathcal{U} = \{b(x-1) + c(x^2-1) \mid b, c \in \mathbb{R}\} = \text{span}\{x-1, x^2-1\}$, so $\{x-1, x^2-1\}$ is a spanning set for $\mathcal{U}$. This set is also independent because it consists of two nonparallel vectors. In conclusion:

$$\{x-1, x^2-1\} \text{ is a basis of } \mathcal{U} \text{ and } \dim \mathcal{U} = 2$$

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$\mathcal{U} = \left\{A \in M_{22} \mid A \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\right\} \subseteq M_{22}$. We look for a basis of $\mathcal{U}$.

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then

$$A \in \mathcal{U} \Leftrightarrow A \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} a-b & 0 \\ c-d & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Leftrightarrow \begin{cases} a=b \\ c=d \end{cases} \Leftrightarrow A = \begin{bmatrix} a & a \\ c & c \end{bmatrix}$$

This means that

$$\mathcal{U} = \left\{ \begin{bmatrix} a & a \\ c & c \end{bmatrix} \mid a, c \in \mathbb{R} \right\} = \left\{ a \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \mid a, c \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\},$$

so $\left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\}$ is a spanning set of $\mathcal{U}$. It is also independent because it is a set formed by two nonparallel vectors. We conclude:

$$\left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\} \text{ is a basis of } \mathcal{U} \text{ and } \dim \mathcal{U} = 2.$$

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a) FALSE: $\{1, x, 2x, 3x\}$ is a set of four nonzero polynomials but it is not a basis because (for instance) it is dependent:

$$0 \cdot 1 + 2 \cdot x + (-1) \cdot 2x + 0 \cdot 3x = 0.$$

b) TRUE: Suppose $a\vec{u} + b(\vec{u} + \vec{v}) = \vec{0}$ ($a, b \in \mathbb{R}$). Then $(a+b)\vec{u} + b\vec{v} = \vec{0}$, and since $\{\vec{u}, \vec{v}\}$ is independent we get $a+b = b = 0$.

This forces $a = b = 0$, so $\{\vec{u}, \vec{u} + \vec{v}\}$ is independent.

c) FALSE: $\{\vec{u}, \vec{v}, \vec{u} + \vec{v}\}$ is dependent because (for instance) $1 \cdot \vec{u} + 1 \cdot \vec{v} + (-1) \cdot (\vec{u} + \vec{v}) = \vec{0}$.

d) TRUE: Suppose $a\vec{u} + b\vec{v} = \vec{0}$ for some $a, b \in \mathbb{R}$. Then $a\vec{u} + b\vec{v} + 0\vec{w} = \vec{0}$, and because $\{\vec{u}, \vec{v}, \vec{w}\}$ is independent we deduce $a = b = 0$.

e) TRUE: If $\{\vec{v}_1, \dots, \vec{v}_n\} \subseteq V$ is independent and $\{\vec{u}_1, \dots, \vec{u}_n\}$ is a basis of V , then by the Steinitz exchange lemma we have

$k \leq n = \dim V$, so an independent subset of V cannot have more than n vectors.